

SmartCart - A Dictionary-Based Fuzzy Search and Spell Correction Engine for Smart Shopping

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Online shopping platforms handle thousands of product searches every minute, and two problems repeatedly affect the customer experience: customers filtering a large catalog by several keywords at once, and customers misspelling product or brand names while searching. Exact string matching cannot handle either case efficiently or gracefully. This project presents SmartCart, a product search and correction engine that combines multiple string-matching and edit-distance algorithms to solve both problems within a single, realistic e-commerce scenario.

The system stores a dictionary of catalog keywords and brand terms in a Trie, organizing them character by character so that words sharing common prefixes share the same path. The Aho-Corasick algorithm is built on top of this Trie using failure links, allowing the system to scan a product description or a customer's multi-keyword filter query once and instantly report every catalog keyword present in it. The KMP algorithm is used when a customer wants to search for one exact product phrase within a description, using the Longest Prefix Suffix (LPS) array to avoid re-scanning characters after a mismatch.

For spelling correction, the system uses the Wagner-Fischer algorithm to compute the standard edit distance between a mistyped search term and each dictionary entry, and the Damerau-Levenshtein algorithm to additionally account for adjacent character transpositions, a very common mobile-typing error. By combining a Trie, Aho-Corasick, KMP, and dynamic-programming-based edit distance techniques within one shopping search assistant, the project demonstrates how these data structures and algorithms work together to solve genuine, everyday retrieval problems.

SCENARIO

SmartCart is a customer-facing search assistant for an e-commerce application. It enables customers to search products using multiple keywords, exact phrases, and misspelled product or brand names. The system applies string-matching and edit-distance algorithms to provide relevant search results and spelling corrections efficiently.

DESCRIPTION

Problem

An e-commerce application must provide fast and flexible product searching. Exact-match search becomes ineffective when customers use multiple keywords or make spelling mistakes. SmartCart addresses these issues by supporting multi-keyword searching, exact phrase lookup, and typo-tolerant search within the same application.

Approach

SmartCart combines a Trie, string-matching algorithms, and edit-distance algorithms to process customer search input against a product dictionary and catalog.

1. Trie

The Trie stores catalog keywords character by character and provides an organized structure for storing and retrieving product terms. It also forms the base structure for Aho-Corasick.

2. Aho-Corasick Exact Matching

Aho-Corasick searches multiple keywords simultaneously within product descriptions using failure links for efficient matching. It supports SmartCart's multi-keyword product filtering.

3. KMP Pattern Searching

KMP is used for exact phrase searching within product descriptions. It uses the LPS array to reduce unnecessary comparisons and perform efficient pattern matching.

4. Wagner-Fischer Edit Distance

Wagner-Fischer calculates the minimum number of insertions, deletions, and substitutions required to transform a mistyped word into a dictionary word. It is used to generate spelling suggestions.

5. Damerau-Levenshtein Edit Distance

Damerau-Levenshtein extends edit distance by considering adjacent character transpositions, allowing SmartCart to handle common typing mistakes more effectively.

Dictionary and Catalog Design

SmartCart maintains two separate sources. The dictionary contains 948 unique shopping terms across 25 categories and is used for Trie storage and typo correction. The catalog contains realistic product descriptions created from dictionary terms and is searched using Aho-Corasick and KMP.

Search and Correction Flow

For multi-keyword searches, Aho-Corasick scans each product description and identifies matching keywords. For exact phrase searches, KMP searches each catalog entry and reports the matching position. For misspelled terms, Wagner-Fischer and Damerau-Levenshtein compare the input with dictionary words and provide the closest correction suggestion.

Outcome

SmartCart provides a practical demonstration of multi-keyword searching, exact phrase matching, and spelling correction using five algorithms from the syllabus. It uses Aho-Corasick to identify multiple keywords, KMP for exact phrase searching, and Wagner-Fischer and Damerau-Levenshtein for typo detection and correction. Supported by a 1753-word shopping dictionary, the project demonstrates how different data structures and string-processing algorithms can be combined to create an efficient e-commerce search system.